MIL-STD-408A
8 December 1960

SUPERSEDING
MIL-STD-408
28 June 1957
MIL-T-0015659E
(SHIPS)
13 June 1958

MILITARY STANDARD

BRUSHES, ELECTRICAL CONTACT RECTANGULAR, STANDARD DESIGNS AND DIMENSIONS FOR

UNITED STATES
GOVERNMENT PRINTING OFFICE
WASHINGTON : 1960

FSC 5977

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T22:02Z
Check the source to verify that this is the current version before use.

ARMED FORCES SUPPLY SUPPORT CENTER
Washington 25, D.C.

Brushes, Electrical Contact, Rectangular,
Standard Designs and Dimensions for
MIL-STD-408A

1. This standard has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force effective.

2. In accordance with established procedure, the Corps of Engineers, Bureau of Ships, and Air Force, respectively, have been designated Army-Navy-Air Force custodians of this standard.

3. Recommended corrections, additions, or deletions should be addressed to the Standardization Division, Armed Forces Supply Support Center, Washington 25, D. C.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T22:02Z
Check the source to verify that this is the current version before use.

INTRODUCTION

(a) This standard corresponds in general, but not in detail, to the U. S. Department of Commerce Simplified Practice Recommendation R56–35, Standard MS91000, ASA Publication No. C64. 1-1956 and NEMA No. CB 1-1956. In addition to being used for the design of brushes it is also intended to hold to a minimum the number of designs and sizes of brush holders.

(b) The brush form designations appearing in this standard are based on interchangeability. For example, the brush form designation remains the same regardless of whether the terminal lug is type S (slot) or H (hole) as long as the terminal lug is suitable for a screw of some certain diameter.

(c) This standard does not cover grades of brushes or tests. Top and bottom bevels, however, which appear in this standard are based on coefficients of friction not exceeding 0.50 for electrographitic grades, 0.45 for graphitic grades and 0.40 for copper-graphitic grades.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T22:02Z
Check the source to verify that this is the current version before use.

CONTENTS

Page

1. SCOPE
1.1 SCOPE
1.2 CLASSIFICATION
2. REFERENCED DOCUMENTS
2.1 SPECIFICATIONS
3. GENERAL REQUIREMENTS
3.1 DEFINITIONS
3.2 STANDARD STYLES
4. DETAIL REQUIREMENTS
4.1 TYPE I, RECTANGULAR, BRUSHES WITH SHUNTS BUT WITHOUT SPRINGS
4.2 TYPE II, RECTANGULAR, WITH SHUNTS AND SPRINGS
4.3 TYPE III, RECTANGULAR, WITH SPRINGS BUT WITHOUT SHUNTS

FIGURES

Figure

1 Bevels
2 Commutator brush
3 Collector ring brush
4 Standard hammer clip
5 Offset hammer clip
6 Extra wide hammer clip
7 Angular hammer clip
8 Single shoulder
9 Double shoulder
10 Beveled edge
11 Beveled corner
12 Terminal lugs
13 Marking
14 Representative style C, commutator brush
15 Representative style R, collector ring brush
16 Terminal, round type, stamped
17 Terminal, round type, machined
18 Terminal, eared type

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T22:02Z
Check the source to verify that this is the current version before use.

CONTENTS

TABLES

		Page
Table I	Dimensional tolerances for rectangular cross-section brushes....	
Table II	Width of toe of beveled brushes.....	
Table III	Depth of chamfer	
Table IV	Shunts for brushes	
Table V	Designations and dimensions applicable to terminal lugs.....	
Table VI	Standard sizes of style C, commutator brushes.....	
Table VII	Standard sizes of style R, collector ring brushes.....	
Table VIII	Dimensional tolerances for rectangular and square cross-section brushes	
Table IX	Dimensional tolerances for brush heads.....	
Table X	Shunts for brushes	
Table XI	Stamped, round type terminals	
Table XII	Machined, round type terminals	
Table XIII	Eared type terminals	
Table XIV	Standard sizes of brushes having both shunts and springs.....	
Table XV	Shunts for standard rectangular brushes with springs.....	

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T22:02Z
Check the source to verify that this is the current version before use.

1. SCOPE

1.1 Scope. This standard covers the standard designs and dimensions for rectangular brushes exclusive of those for aero-space applications.

1.2 Classification.

1.2.1 This standard covers the following types and styles of brushes:

Type I—Rectangular, with shunts but without springs.

Style C—Commutator.

Style R—Collector ring.

Type II—Rectangular, with shunts and springs.

Type III—Rectangular, with springs but without shunts.

Type IV—Rectangular, with neither shunts nor springs.

2. REFERENCED DOCUMENTS

2.1 Not applicable.

3. GENERAL REQUIREMENTS

3.1 Definitions. The following definitions are applicable to this standard:

3.1.1 *Length.* The length of a brush is the maximum overall dimension in the direction in which the brush feeds to the commutator or collector ring. This is also known as the radial dimension.

3.1.2 *Width.* The width of a brush is the dimension at right angles to the length and to the direction of rotation. This is also known as the axial dimension.

3.1.3 *Thickness.* The thickness of a brush is the dimension at right angles to the length in the direction of rotation. This is also known as the tangential dimension.

3.1.4 *Front side.* The front side of a brush having one or both ends beveled is the area bounded by the width and the shortest length.

3.1.5 *Back side.* The back side of a brush having one or both ends beveled is the area bounded by the width and the greatest length.

3.1.6 *Bevels and toes.* The top bevel and bottom bevel are the angles included between the beveled surface and a plane at right angles to the length. The toe is the uncut or flat portion at the beveled end (see figure 1).

3.1.7 *Split brush.* A split brush is one which consists of two or more pieces used in place of a single brush. The adjacent sides of the split brush pieces are parallel to the commutator slots.

3.1.8 *Paired brush.* A paired brush is one which consists of two individual brushes which are joined together by a common shunt or terminal lug. A paired brush is not a split brush.

3.1.9 *Headed brush.* A headed brush is one having a top smaller in cross-section than the width times the thickness. The head is usually cylindrical or conical in shape.

3.1.10 *Pressure condition 1.* Pressure condition 1 is the so-called "new-brush" condition on the complete brush assembly.

3.1.11 *Pressure condition 2.* Pressure condition 2 is the condition after the brush has undergone 100 percent wear.

3.1.12 *Brush face.* Brush face is the end of the brush which makes contact with commutator or collector ring.

3.2 Standard styles.

3.2.1 *Type I.* Standard styles for style C (commutator) brushes smaller than 1½ inches long by ¾ inch wide by ⅜ inch thick and for

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T22:02Z
Check the source to verify that this is the current version before use.

style R (collector ring) brushes smaller than 1½ inches long by ⅜ inch wide by ¾ inch thick will be established by revision of this standard. The requirements to be specified shall correspond with those specified in 4.1.13, 4.1.14, 4.1.15 and 4.1.16 herein.

3.2.2 *Type II.* Standard styles for type II brushes smaller than $\frac{3}{16}$ inch thick by ¼ inch wide by ½ inch long will be established by revision of this standard. The following will be covered:

Brush:
- Length (see 3.1.1).
- Width (see 3.1.2)
- Thickness (see 3.1.3).

Shunt:
- Designation (see 4.2.3).

Terminal:
- Type (see 4.2.4.1).
- Dimensions (see 4.2.4.2).

Spring:
- Minimum permissible pressure in ounces under both pressure conditions 1 and 2.
- Maximum permissible pressure in ounces under pressure condition 1.
- Compressed length in inches under both pressure conditions 1 and 2.

3.2.3 *Type III.* Standard styles for type III brushes smaller than $\frac{3}{16}$ inch thick by ¼ inch wide by ½ long will be established by revision of this standard. The following will be covered:

Brush:
- Length (see 3.1.1).
- Width (see 3.1.2).
- Thickness (see 3.1.3).

Spring:
- Minimum permissible pressure in ounces under both pressure conditions 1 and 2.
- Maximum permissible pressure in ounces under pressure condition 1.
- Compressed length in inches under both pressure conditions 1 and 2.

3.2.4 *Type IV.* Standard styles for type IV will be established by revision of this standard. The following will be covered:

Brush:
- Length (see 3.1.1).
- Width (see 3.1.2).
- Thickness (see 3.1.3).

4. DETAIL REQUIREMENTS

4.1 Type I, rectangular, with shunts but without springs.

4.1.1 *Shunts locations and connections.* The locations by position number shall be identified as shown on figures 2 and 3. For all brushes on which the shunt is fastened by means of tubular rivets, the rivets shall be so spun that when a well rounded head is produced on both sides it shall not protrude beyond the sides of the brush. Rivets shall be made of materials which do not distort under conditions of assembly or operation, to the extent that the connection is weakened. Care shall be taken that the soldering does not affect the flexibility of the shunt or shunts. Riveted connections shall be mechanically and electrically secure to prevent movement or excessive change in millivolt drop due to vibration and shock of operation.

4.1.2 *Hammer clips.* Hammer clips shall be identified as shown on figures 4 to 7, inclusive. Unless corrosion-resistant material is used, all hammer or guide clips shall be tinned after blanking or bending. Tinning shall be accomplished by electrodepositing to a minimum thickness of 0.0005 inch or by a hotdip process. The hot-dip process shall be such as to insure uniform thickness, and shall not damage the base metal. The gage of the material shall be such that the clip will not fracture or break under any normal conditions of operation. The inside bend radius shall be not less than 2 times the thickness of the metal. The cutting of ears, such as those which

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T22:02Z
Check the source to verify that this is the current version before use.

TABLE I. *Dimensional tolerances for rectangular cross-section brushes.*

	Electrographitic and graphitic grades		Copper-graphic grades containing 50 percent or more metal by weight		
	Plus	Minus	Plus	Minus	Minus
	Inch	*Inch*	*Inch*	*Inch*	*Inch*
Length	0.032	0.032	0.032	0.032	.....
Width ¾ inch or under	.000	.004	.000	.004	.....
Width over ¾ inch	.000	.015	.....	.007	0.020
Thickness	.000	.004	.....	.....	.....
Thickness over ½ inch	.....	.....	.....	.007	.011
Thickness ½ inch and under	.....	.....	.....	.000	.004

have been used in certain designs for retaining the brush under shock, in the bend of the hammer clip will not be permitted.

4.1.3 *Shoulders.* Shoulders shall be single or double as specified and shall be identified as shown on figures 8 and 9.

4.1.4 *Beveled edges and corners.* Beveled edges and corners (as distinguished from top and bottom bevels) shall be identified as shown on figures 10 and 11.

4.1.5 *Dimensional tolerances.* Where lengths of 1½, 2, and 2¼ inches are specified herein, lengths of 1¾, 2¼, 2½ inches respectively may be utilized provided the machinery application and brush holder design are such that replacement may be accomplished with a brush of the length specified herein. In any case, the dimensional tolerances shall not exceed those shown on table I, except as follows: Where length and width are the same, width tolerances apply to both length and width. For split brushes, the variation in thickness shall be the same as for an equivalent solid brush.

4.1.6 *Bevel tolerances.* Bevel angles shall be accurate to within 1 degree above or below the nominal angle specified herein.

4.1.7 *Toes.* The width of the toe of beveled brushes shall be as shown in table II.

TABLE II. *Width of toe of beveled brushes.*

Brush thickness	Maximum width of toe
Inch	*Inch*
Up to and including ¾	1/16
Over ¾	⅛

4.1.8 *Chamfers.*

4.1.8.1 *Angle of chamfer.* The angle of chamfer shall be such that an equal amount of material is taken from the two adjacent sides. The angle between the plane of the side of the brush and the plane of the face of the chamfer extended shall be 45 degrees as measured with a protractor.

4.1.8.2 *Depth of chamfer.* The depth of chamfer on each surface shall be measured along the plane on each of the two adjacent surfaces, and not the width of the face of the cut portion. The standard depth of chamfer of brushes shall be as shown in table III.

TABLE III. *Depth of chamfer.*

Thickness of brush (any width)	Depth of chamfer	Tolerances	
		Plus	Minus
Inch	*Inch*	*Inch*	*Inch*
¼ or less	0.010	0.005	0.000
Over ¼, but less than ⅜	.032	.016	.000
⅜ to ⅝, inclusive	.062	.032	.000
Over ⅝	.093	.032	.000

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T22:02Z
Check the source to verify that this is the current version before use.

TABLE IV. *Shunts for brushes.*

Designation used in Military standard sheets	Area of cross section, nominal	Diameter of strands, nominal	Number of strands in each member	Number of members
	Circular mils	*Mils*		
A	132	2	11	3
B	264	2	22	3
C	326	3	11	3
D	652	3	22	3
E	1,038	3	15	7
F	1,575	5	9	7
G	2,625	5	15	7
H	4,200	5	24	7
J	6,475	5	37	7
K	10,325	5	59	7
L	13,125	5	75	7
M	16,625	5	95	7
N	20,825	5	119	7
P	26,250	5	150	7

4.1.9 *Shunts.*

4.1.9.1 *Lengths.* The length of the shunts shall be the distance from the extreme top of the brush to the center of the hole or round portion of the slot in the terminal lug or the center of the inserted portion of the plug terminal. In determining the length of shunt, the shunt shall be extended in in a direction parallel with the length dimension of the brush. The length shall be 1, 1⅝, 2¼, 3, 4, 5½ or 7½ inches. Tolerances for

TABLE V. *Designations and dimensions applicable to terminal lugs.*

Designation	Designation	Dimensions inches			Maximum [1] conductor size, circular mils	Thickness range inches
		A	B	C		
FH040812	FS040812	1/8	1/4	3/8	1575	0.020—0.025
FH050913	FS050913	5/32	9/32	13/32	2625	.020— .025
FH051216	FS051216	5/32	3/8	3/8	6475	.025— .030
FH071218	FS071218	7/32	3/8	9/16	6475	.025— .030
FH071624	FS071624	7/32	1/2	1/2	20825	.030— .040
FH092028	FS092028	9/32	5/8	7/8	26250	.040— .046
H040916	S040916	1/8	9/32	1/2	6475	.020— .025
H051016	S051016	5/32	5/16	1/2	6475	.020— .030
H051020	S051020	5/32	5/16	5/8	10325	.020— .030
H061220	S061220	3/16	3/8	5/8	26250	.020— .030
H071428	S071428	7/32	7/16	7/8	26250	.030— .040
H091832	S091832	9/32	9/16	1	52500	.030— .040
H112236	S112236	11/32	11/16	1 1/8	52500	.040— .046
H132640	S132640	13	13/16	1 1/4	52500	.040— .046
TH091832	TS091832	9/32	9/16	1	52500	.....
TH112232	TS112232	11/32	11/16	1	52500	.....
TH132632	TS132632	13/32	13/16	1	52500	.....

[1] These conductors are of the stranded type consisting of seven members, each composed strands 5 mils in diameter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T22:02Z
Check the source to verify that this is the current version before use.

the first three shall be plus 1/8 inch minus 0 inch. Tolerances for the last four lengths shall be plus 1/4 inch minus 0 inch.

4.1.9.2 *Sizes.* Shunts shall conform to table IV.

4.1.10 *Terminal lugs.*

4.1.10.1 *Types.* The terminal lugs shall conform with one of the designs shown on figure 12.

4.1.10.2 *Sizes.* The sizes shall conform with table V.

4.1.11 *Marking.* Where space permits, the manufacturer's grade designation shall be placed above the point of 100 percent wear (see fig. 13) so that the information is available when the brush has been worn to that point.

4.1.11.1 *One hundred percent wear.* For the purposes of this standard, the point of 100 percent wear shall be as follows:

Brushes with shunts but without springs:

Brushes with riveted connections: A point 1/16 inch from the metallic part nearest the brush face and in the direction toward the brush face.

Brushes with embedded connections: If with hammer clip, at a point 1/16 inch from the metallic part nearest the brush face and in the direction toward the brush face. If without hammer clip, at a distance from the brush face equal to 0.55 the original length of the brush.

Brushes with shunts and springs: At a distance from the brush face equal to 0.65 multiplied by the remainder obtained by subtracting the original length of the head from the total length of the brush.

4.1.11.2 *Sizes 1/2 inch thick by 3/4 inch wide by 3/4 inch long, or less.* Each brush shall be marked with the following information:

(a) Manufacturer's grade designation.

(b) Point of 75 percent wear. The marking shall be permanent and legible and may be accomplished by machining, stamping or other suitable means. The preferred method is shown on figure 12. Another satisfactory method is a straight line instead of a "V". Straight wear lines running diagonally across either the width or the thickness of the brush material fulfill these requirements.

4.1.11.3 *Sizes larger than 1/2 inch thick by 3/4 inch wide by 3/4 inch long.* Each brush shall be provided with the following information:

(a) Manufacturer's grade designation.

(b) Military grade designation. Marking shall be by metal stamp, engraving or a similar process that indents or recesses letters and symbols.

4.1.12 *Style designations.* The basis for developing style designations herein is as follows:

(a) *Size.*—The thickness, width, and length are expressed in sixteenths of an inch and listed in the order of thickness, width, and length. In a split brush the thickness dimension is the total thickness of the brush. In a paired brush, the dimensions, given are for each individual brush. The series is followed by a "C" if the brush is for use on a commutator and "R" if the brush is for use on a collector ring.

(b) *Bevels.* — (For commutator brushes only.) The numerals 0, 1, 2, 3, 4, 5, 6, 7 and 8 represent bevels of 0, 5, 10, 15, 20, 25, 30, 35 and 40 degrees, respectively. The top bevel is given first and the bottom bevel second.

(c) *Example.*—A style designation 082036-C87, indicates, a brush, 1/2 inch thick, 1 1/4 inch wide and 2 1/4 inch long, having a top bevel of 40 degrees and a bottom bevel of 35 degrees, and suitable for commutators.

4.1.13 *Standard sizes of style C commutator brush.*—Commutator brushes shall conform to one of the sizes shown in table VI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T22:02Z
Check the source to verify that this is the current version before use.

TABLE VI. *Standard sizes of style C, commutator brushes.*

Designation	Thickness	Width	Length	Shunt designation (see 4. 1. 9. 2) where brush current density is 60 amp./sq. in. or less	Shunt designation (see 4. 1. 9. 2) where brush current density is 100 amp./sq. in. or less but greater than 60 amp./sq. in.
	Inch	*Inches*	*Inches*		
061224	3/8	3/4	1 1/2	One H or two F	One J or two G
061624	3/8	1	1 1/2	One H or two G	One K or two H
081624	1/2	1	1 1/2	One J or two G	One L or two J
101624	5/8	1	1 1/2	One K or two H	One N or two J
121624	3/4	1	1 1/2	One L or two H	One P or two K
062032	3/8	1 1/4	2	One J or two G	One L or two J
082032	1/2	1 1/4	2	One K or two H	One N or two J
102032	5/8	1 1/4	2	One L or two H	One P or two K
122032	3/4	1 1/4	2	One M or two J	Two L
162032	1	1 1/4	2	One P or two K	Two N
062436	3/8	1 1/2	2 1/4	One K or two H	One M or two J
082436	1/2	1 1/2	2 1/4	One L or two H	One P or two K
102436	5/8	1 1/2	2 1/4	One M or two J	Two L
082840	1/2	1 3/4	2 1/2	One M or two J	Two K
102840	5/8	1 3/4	2 1/2	One N or two J	Two M

4.1.14 *Standard sizes of style R, collector ring brush.* Style R collector ring brushes shown on figure 15 shall conform to one of the sizes shown in table VII.

TABLE VII. *Standard sizes of style R, collector ring brushes.*

Designation	Thickness	Width	Length	Shunt length	Shunt designation (see 4. 1. 9. 2) where brush current density is 60 amp./sq. in. or less	Shunt designation (see 4. 1. 9. 2) where brush current density is 100 amp./sq. in. or less but greater than 60 amp./sq. in.
	Inches	*Inch*	*Inches*	*Inches*		
120624	3/4	3/8	1 1/2	3	One H or two F	One J or two G
100618	5/8	3/8	1 1/8	4	One H or two F	One J or two G
120820	3/4	1/2	1 1/4	4	One H or two G	One K or two H
160624	1	3/8	1 1/2	4	One H or two G	One K or two H
161024	1	5/8	1 1/2	4	One K or two H	One N or two J
201232	1 1/4	3/4	2	4	One M or two J	Two L

4.1.15 *Style C, commutator brush.* A representative style C commutator brush is shown on figure 14.

4.1.15.1 *Bevels.* Bevels shall be as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T22:02Z
Check the source to verify that this is the current version before use.

	Top bevel	Bottom bevel	Bevel designation
	Degrees	Degrees	
Radial brushes for reversing machines	0	0	00
Radial brushes for nonreversing machines	30	0	60
Trailing brushes for nonreversing machines	15	15	33
Leading brushes for nonreversing machines	[1]	[2]	..

[1] The preferred angle is 40 degrees. Angles of 35 and 30 degrees are permissible, however.

[2] The preferred angle is 35 degrees. Angles of 30 and 40 degrees are permissible, however. In the interest of reducing the number of different items of supply, the stocking of brushes with a 0-degree bottom bevel is encouraged; in this case the required bottom bevel is obtained by cutting and sanding when the brushes are installed in a machine for which they are intended.

4.1.15.2 *Shunt locations.* Shunt locations shall be positions 1, 2, 3, 7, 8 or 9, 10 or 12 (see 4.1.1). Where the design includes an angular hammer clip (see 4.1.15.3) and a double, one-piece, loop-type shunt, the cut ends of which are secured to the terminal lug or lugs, contact may be made by soldering the shunt to the notch of the hammer clip.

4.1.15.3 *Hammer clips.* Hammer clips shall be standard, offset, extra wide or angular (see 4.1.2).

4.1.15.4 *Shoulders.* Brushes may be furnished without shoulders or with single or double shoulders (see 4.1.3).

4.1.15.5 *Beveled edges and corners.* Brushes may be furnished either with beveled edges or corners as shown on figures 11 and 12, respectively (see 4.1.4), or without beveled edges or corners.

4.1.15.6 *Shunt connection.* The connection shall be riveted or tamped. Shunts are not required to be insulated. Where two shunts are supplied on one brush, both shunts shall be attached to the same terminal lug.

4.1.15.7 *Shunt length.* Shunt length of brushes from 1½ by ¾ by ⅜ inches to 1½ by 1 by ⅝ inches, inclusive, shall be 3 inches. Shunt length of brushes from 2 by 1 by ¾ inches to 2½ by 1¾ by ⅝ inches, inclusive, shall be 4 inches (see 4.1.9.1).

4.1.16 *Style R collector ring brush.*

4.1.16.1 A representative style R collector ring brush is shown on figure 15.

4.1.16.2 *Contact curvature.* The contact curvature shall be suitable for a collector ring having a diameter equal to ten times the brush thickness.

4.1.16.3 *Shunt locations.* Shunt locations shall be positions 1, 4, 7, 11, 3, 6, 9 or 13 (see 4.1.1).

4.1.16.4 *Hammer clips.* Hammer clips shall be standard, offset, or extra wide (see 4.1.2).

4.1.16.5 *Shoulders.* Brushes may be furnished without shoulders or with single or double shoulders (see 4.1.3).

4.1.16.6 *Beveled edges and corners.* Brushes may be furnished either with beveled edges or corners as shown on figures 11 and 12, respectively (see 4.1.4), or without beveled edges or corners.

4.1.16.7 *Shunt connection.* The connection shall be riveted or tamped. Shunts are not required to be insulated. Where two shunts are supplied on one brush, both shunts shall be attached to the same terminal lug.

4.2 Type II, rectangular, with shunts and springs.

4.2.1 *Dimensional tolerances.* The dimensional tolerances shall not exceed those specified in table VIII except as follows: Where thickness is greater than width, the thickness tolerance shall apply to both width and thickness. Where length and width are the same, width tolerances apply to both length and width.

4.2.1.1 *Brush heads.* The dimensional tolerances for brush heads shall not exceed those specified in table IX.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T22:02Z
Check the source to verify that this is the current version before use.

TABLE VIII. *Dimensional tolerances for rectangular and square cross-section brushes.*

	Electrographitic and graphitic grades		Copper-graphite grades containing 50 percent or more metal by weight		
	Plus	Minus	Plus	Minus	Minus
	Inch	*Inch*	*Inch*	*Inch*	*Inch*
Length	0.032	0.032	0.032	0.032	.....
Width ¾ inch or under	.000	.004	.000	.004	.....
Width over ¾ inch	.000	.015	.....	.007	0.020
Thickness	.000	.004	.....	.....	.....
Thickness over ½ inch	.....	.....	.....	.007	.011
Thickness ½ inch and under	.....	.....	.....	.000	.004
Head length	.016	.000	.016	.000	.....

TABLE IX. *Dimensional tolerances for brush heads.*

	Tolerance	
	Plus	Minus
	Inch	*Inch*
Diameter	0.000	0.006
Length	1/64	.000
Location	Center of head shall not vary more than 0.005 inch from center of brush at the plane of the shoulder	

TABLE X. *Shunts for brushes.*

Area of cross section, nominal	Diameter of strands, nominal	Number of strands in each member	Number of members
Circular mils	*Mils*		
72	2	18	1
100	2	25	1
132	2	11	3
172	2	6	3
204	2	17	3
264	2	22	3
326	3	33	1
420	2	15	7
600	2	50	3
1120	2	40	7
2016	2	72	7
3220	2	115	7
296	3	15	2
326	3	11	3
444	3	15	3
652	3	22	3
740	3	11	7
888	3	13	7
1038	3	15	7
1869	3	27	7
4568	3	66	7
6900	3	100	7

4.2.2 *Chamfers.*

4.2.2.1 *Angle of chamfer.* The angle of chamfer shall be such that an equal amount of material is taken from the two adjacent sides. The angle between the plane of the side of the brush and the plane of the face of the chamfer extended shall be 45 degrees as measured with a protractor.

4.2.2.2 *Depth of chamfer.* The depth of chamfer on each surface shall be measured along the plane on each of the two adjacent surfaces, and not the width of the face of the cut portion. The standard depth of chamfer of brushes shall be as shown in table III.

4.2.3 *Shunts.* Shunts shall conform to table X. The shunt shall be secured in the top of the brush or brush head opposite the contact surface.

4.2.4 *Terminals.* All terminals shall be hot-tinned or silver plated.

4.2.4.1 *Types.* Terminals shall be as follows (see figures 16, 17 and 18).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T22:02Z
Check the source to verify that this is the current version before use.

Round type, stamped.

Round type, machined.

Eared type.

4.2.4.2 *Dimensions.*

4.2.4.2.1 *Round type, stamped.* The dimensions shall conform to table XI. The terminal shall conform to figure 16.

4.2.4.2.2 *Round type, machined.* The dimensions shall conform to table XII. The terminal shall conform to figure 17.

TABLE XI. *Stamped, round type terimnals.*

Designation	D1	D2	T1
	Inch	*Inch*	*Inch*
0120903	3/16	9/64	3/64
0161105	1/4	11/16	5/16
0201504	5/16	15/64	1/16
0241604	3/8	1/4	1/16
0282207	7/16	11/32	7/64
0262608	9/16	7/16	1/8
0483208	3/4	1/2	1/8

NOTE.—The plan for setting up the designations under table XI is as follows: At the left are the type letters. Each pair of numbers designates the number of sixty-fourths of an inch in the appropriate dimension. The pairs are lined up in the same order as the columns are in the table. These designations correspond with the NEMA full designations in ASA Publication No. C64. 1-1956, NEMA Publication No. CB 1-1956.

TABLE XII. *Machined, round type terminals.*

Designation	D1	D2	T1	T2
	Inch	*Inch*	*Inch*	*Inch*
OM 1411-0603	7/32	11/64	3/32	3/64
OM 1613-0604	1/4	13/64	3/32	1/16
OM 1816-0806	9/32	1/4	1/8	1/16
OM 2018-0804	5/16	9/32	1/8	1/16
OM 2218-0804	11/32	9/32	1/8	1/16
OM 3222-0804	1/2	11/32	1/8	1/16
OM 3626-0804	9/16	13/32	1/8	1/16

4.2.4.2.3 *Eared type.* The dimensions shall conform to table XIII. In no case shall the length of the terminal lug ears exceed the solid compressed length of the spring (see 4.2.5). The terminal shall conform to figure 18.

TABLE XIII. *Eared type terminals.*

Designation	D1	W1	W2	We	Le
	Inch	*Inch*	*Inch*	*Inch*	*Inch*
U161412-0512	1/4	7/32	3/16	5/64	3/16
U181612-0812	9/32	1/4	3/16	1/8	3/16
U221814-1414	11/16	9/32	7/32	7/32	7/32
U302420-1616	15/32	3/8	5/16	1/4	1/4
U383228-1624	19/32	1/2	7/16	1/4	3/8

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T22:02Z
Check the source to verify that this is the current version before use.

TABLE XIV. *Standard sizes of brushes having both shunts and springs.*

Designation	Dimensions of brush material (in 1/16 inch)			Minimum permissible pressure (ounces)		Maximum permissible pressure (ounces)	Compressed length of spring (in 1/16 inch)	
	Thickness	Width	Length	Condition 1	Condition 2	Condition 1	Condition 1	Condition 2
030412	3	4	12	4.0	3.4	4.5	12	18
040512	4	5	12	6.7	5.7	7.5	12	18
040616	4	6	16	8.1	6.9	9.0	12	18
040716	4	7	16	9.4	8.0	10.5	12	18
040816	4	8	16	10.8	9.2	12.0	16	24
041020	4	10	20	13.5	11.5	15.0	20	30

NOTES:

1. The above designation is based on expressing in one-sixteenths of an inch, the thickness, width and length in that order.
2. Condition 1 represents the condition when the brush is new. Condition 2 represents the condition when the brush has undergone 100 percent wear. (For the purposes of this table, the following assumptions were made in deriving the value for condition 2: A length of head equal to 11 percent of the total length of the brush. The point of 100 percent wear being at a distance from the contact surface equal to 0.5 multiplied by the remainder obtained by subtracting the length of the head from the total length of the brush.)
3. The distance between the commutator or collector ring and the bottom of the brush holder cap or bottom of the recess in the cap equal to twice the length of the brush when new.
4. Spring pressure is specified on the following basis:
   a. A maximum pressure of 6.0 pounds per square inch when the brush is new.
   b. A minimum pressure of 5.4 pounds per square inch when the brush is new.
   c. A minimum pressure of 4.8 pounds per square inch when the brush has undergone 100 percent wear.

4.2.4.2.4 *Tolerances.* The tolerances of the dimensions appearing in tables XI, XII and XIII shall be as follows:

Le—plus or minus 1/64 inch.

All others—plus 0.000 inch, minus 0.008 inch.

4.2.5 *Springs.* Springs shall be of the positive gradient, compression, helical wire type. There shall be two closed turns at each end of the spring.

4.2.6 *Marking.* Each brush shall be marked with the following information:

(a) Manufacturers' grade designation. If possible, this marking shall be placed above the point of 100 percent wear so that the information is still available when the brush has worn to that point.

(b) Indication of points of 75 and 100 percent wear. The wear points shall be so marked that there is a continuous marking between the two points of wear along the length of the brush.

4.2.7 *Standard sizes.* Standard sizes for rectangular brushes with shunts and springs shall conform to table XIV. The shunts for these brushes shall conform to table XV.

TABLE XV.

Shunts for standard rectangular brushes with springs.

Brush designation	Minimum permissible shunt size where brush current density is 60 amp./sq. in. or less (see note)	Minimum permissible shunt size where brush current density is 100 amp./sq. in. or less but greater than 60 amp./sq. in. (see note)
030412	7-15-002	3-50-002
040512	7-15-002	7-11-003
040616	3-50-002	7-40-002
040716	7-11-003	7-27-003
040816	7-13-003	7-27-003
041020	7-15-003	7-27-003

NOTE.—In these shunt designations, the first figure represents the number of members, the second figure represents the number of strands in each member, and the third figure represents the nominal diameter of strands in inches.

4.3 Type III, rectangular, with springs but without shunts.

4.3.1 *Dimensions and tolerances.* The requirements of 4.2.1, 4.2.2, and 4.2.5 shall apply.

4.3.2 *Marking.* Each brush shall be marked with the manufacturer's grade designation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T22:02Z
Check the source to verify that this is the current version before use.

4.3.3 *Standard sizes.* Standard sizes for rectangular brushes with springs but without shunts shall conform to table XIV except that sizes 040716, 040816, and 041020 shall not be considered standard for type III.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.

Copies of this standard for Military use may be obtained as indicated in the foreword to the Index of Military Specifications and Standards.

The title and identifying symbol should be stipulated when requesting copies of military standards.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodians:

Army—Signal Corps
Navy—Bureau of Ships

Preparing activity:

Navy—Bureau of Ships
(Project 5977-0003)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T22:02Z
Check the source to verify that this is the current version before use.

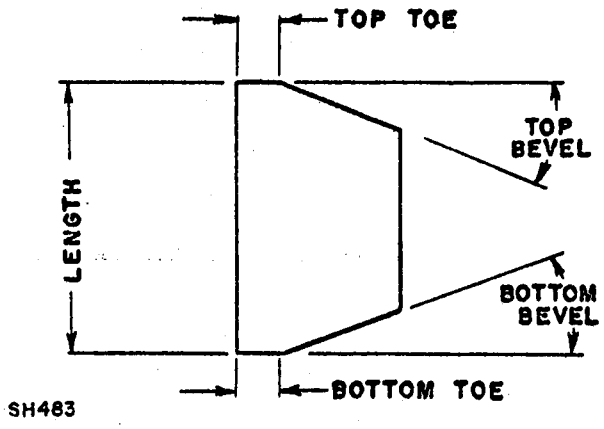


FIGURE 1.—*Bevels.*

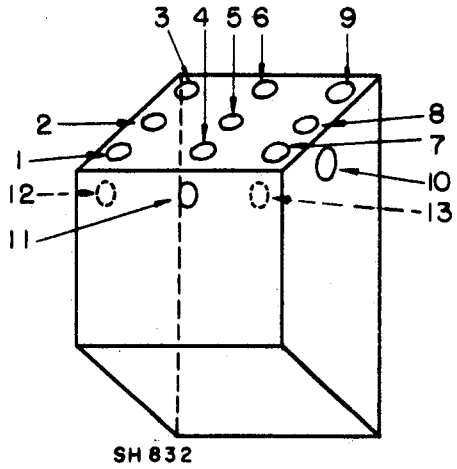


FIGURE 2.—*Commutator brush.*

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T22:02Z
Check the source to verify that this is the current version before use.

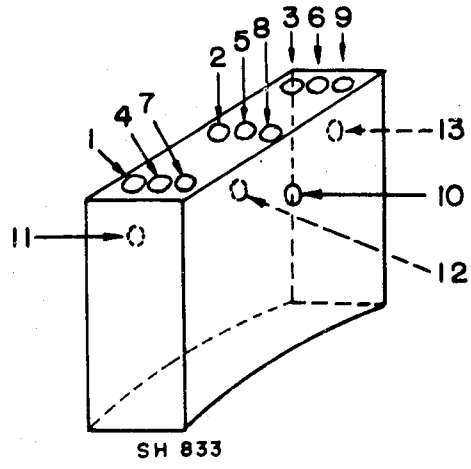


FIGURE 3.—*Collector ring brush.*

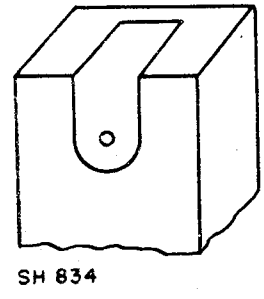


FIGURE 4.—*Standard hammer clip.*

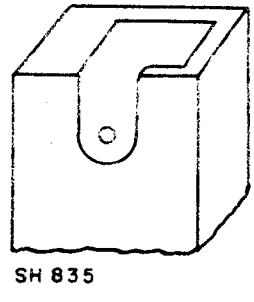


FIGURE 5.—*Offset hammer clip.*

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T22:02Z
Check the source to verify that this is the current version before use.

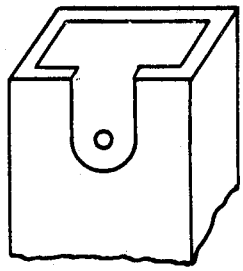


FIGURE 6.—*Extra wide hammer clip.*

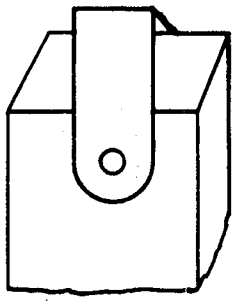


FIGURE 7.—*Angular hammer clip.*

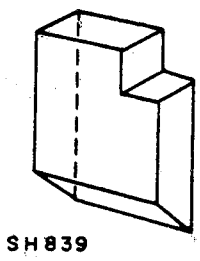


FIGURE 8.—*Single shoulder.*

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T22:02Z
Check the source to verify that this is the current version before use.

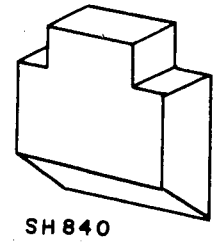


FIGURE 9.—*Double shoulder.*

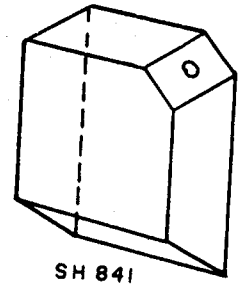


FIGURE 10.—*Beveled edge.*

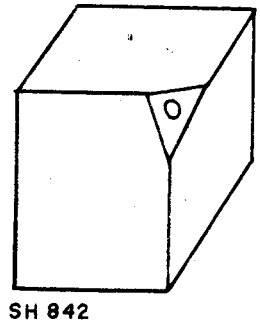
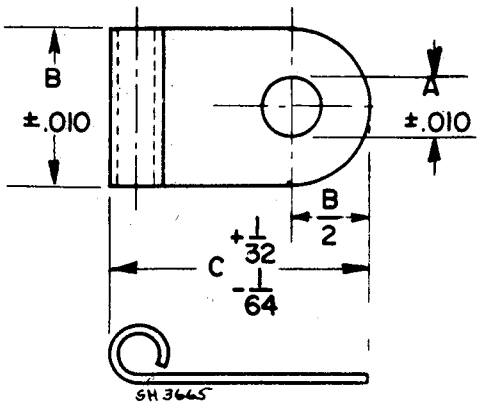
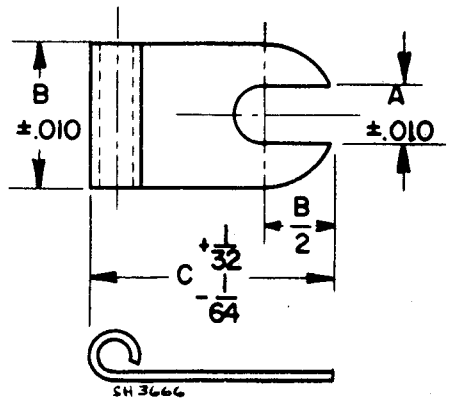


FIGURE 11.—*Beveled corner.*

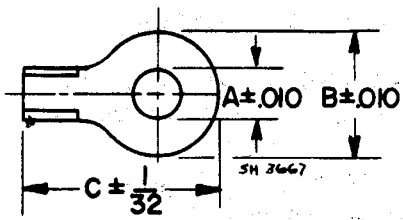
Source: https://assist.dla.mil -- Downloaded: 2026-10-23T22:02Z
Check the source to verify that this is the current version before use.



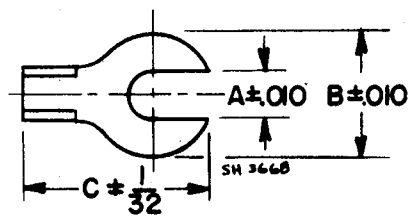
Type FH (flag type with hole).



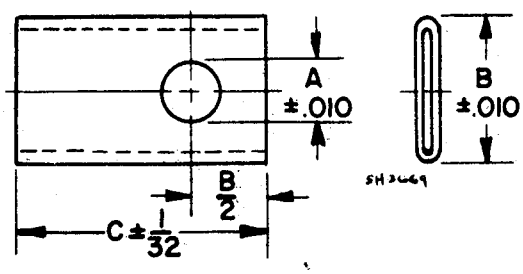
Type FS (flag type with slot).



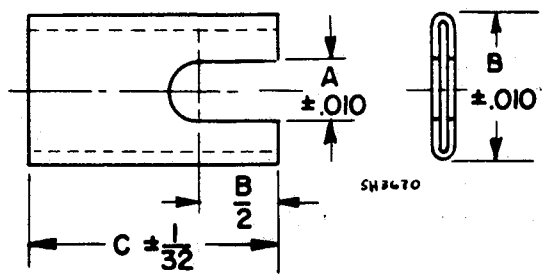
Type H (straight type with hole).



Type S (straight type with slot).



Type TH (tubular type with hole).



Type TS (tubular type with slot).

FIGURE 12.—*Terminal lugs.*

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T22:02Z
Check the source to verify that this is the current version before use.

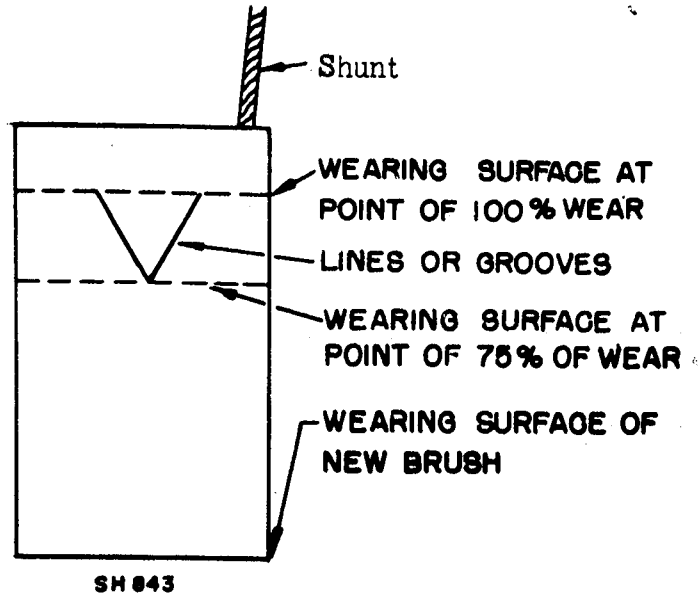


FIGURE 13.—*Marking.*

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T22:02Z
Check the source to verify that this is the current version before use.

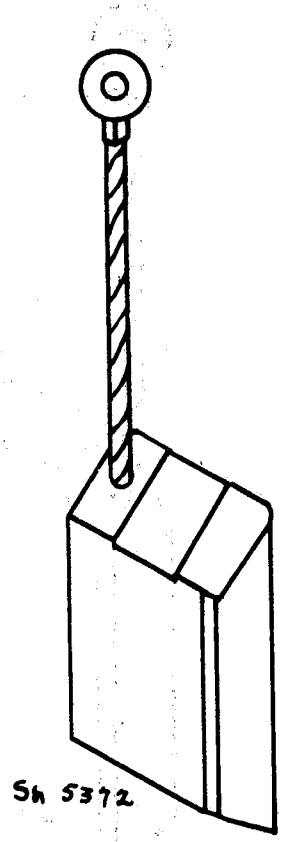


FIGURE 14.—*Representative style C, commutator brush.*

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T22:02Z
Check the source to verify that this is the current version before use.

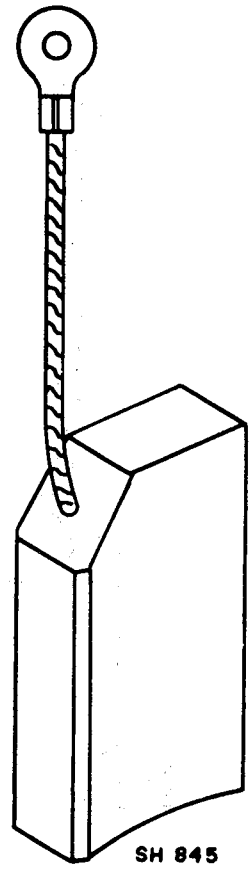


FIGURE 15.—*Representative style R, collector ring brush.*

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T22:02Z
Check the source to verify that this is the current version before use.

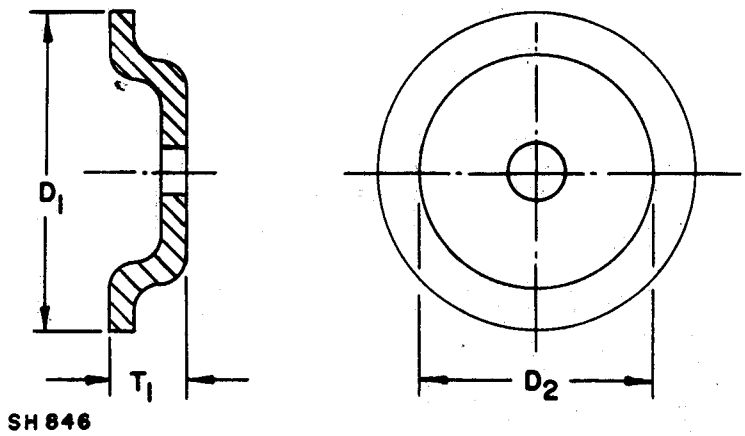


FIGURE 16.—*Terminal, round type, stamped.*

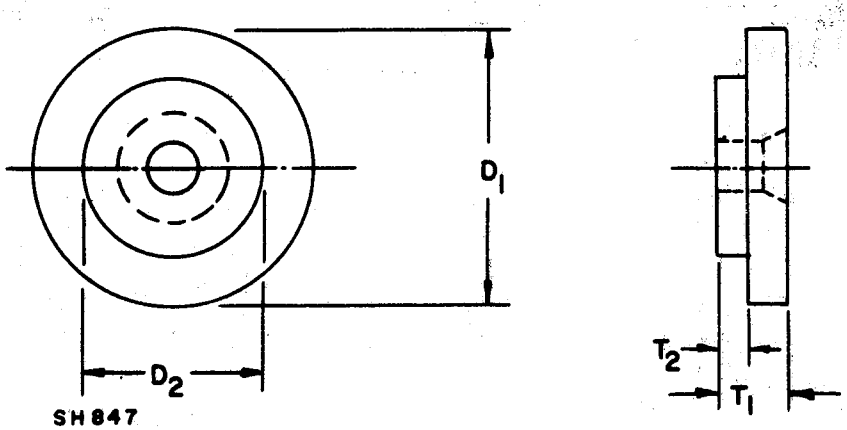


FIGURE 17.—*Terminal, round type, machined.*

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T22:02Z
Check the source to verify that this is the current version before use.

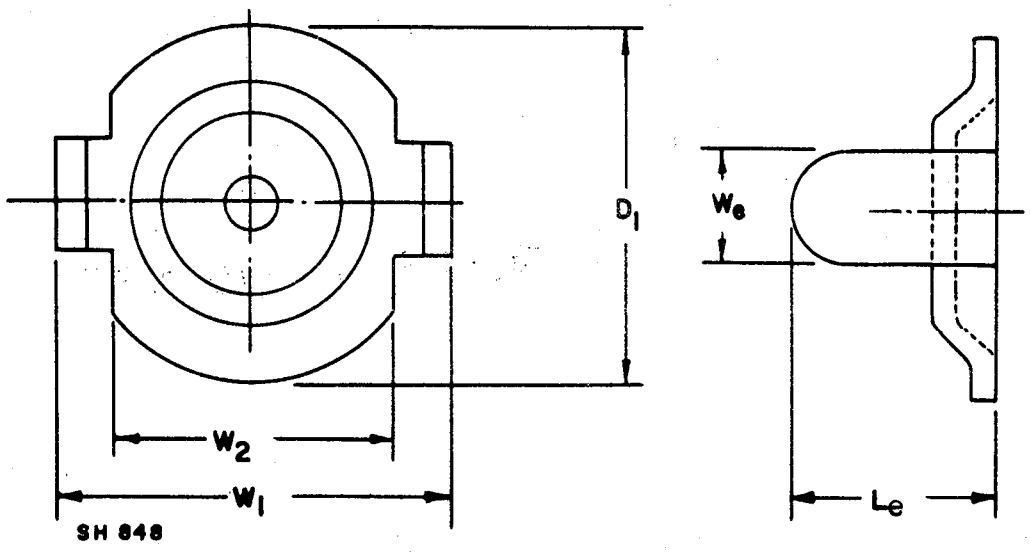


FIGURE 18.—*Terminal, eared type.*

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T22:02Z
Check the source to verify that this is the current version before use.